COMP232 Lab 8

Formal Methods, and ProVerif

The goal of this session is to try working with a modern formal verification tool for security protocols, *ProVerif* by Bruno Blanchet et al. The tool with all related documentation can be found at https://bblanche.gitlabpages.inria.fr/proverif/

We will use the online demo of ProVerif available here: http://proverif24.paris.inria.fr/

> **References:**
>
> - [ProVerif User Manual]()

Task 1: Exploring ProVerif Online Demo

- In a web browser go to the online demo of ProVerif: http://proverif24.paris.inria.fr/
- Load one of the available protocols (Eg: *Needham-Schroeder public key*).
- Observe, in the text box on the right, that the protocol is described using formal syntax of ProVerif. The text enclosed inside `(*` and `*)` are *comments*, and can be easily understood.
- Click *Verify*.
- Click *ProVerif HTML output*, after the previous process is complete.

Some parts of the formal protocol specification can be understood without any preliminary knowledge of ProVerif. For other parts, you can refer to the User Manual linked above.

Task 2: Getting Started with ProVerif

- Go through Chapter 2, Getting Started (pages 7–9) of the User Manual linked earlier.
- Load and execute ProVerif scripts `hello.pv` and `hello_ext.pv` discussed there in ProVerif. The files `hello.pv` and `hello_ext.pv` are available on the same Canvas module.
- Make sure you understand the meaning of the key words `free`, `process` and `query`, and the role they play in the verification process.
- Explain what condition is being checked by the query: `Query event(evCocks) ==> event(evRSA)`.

Task 3: Advanced ProVerif Usage (after the lab optional)

- Go through the Chapter 3, Using ProVerif, follow the proposed steps using Online Demo and handshake protocol specified in Proverif script `ex_handshake.pv`

- Understand all parts of the handshake protocol specification, in particular, the `let` construction used there.
- Understand how the example handshake protocol is verified, and what is the attack found by ProVerif.